

SaaS Support Operations & Customer Satisfaction Analysis

Business Problem: How can a SaaS support team identify operational bottlenecks, improve ticket resolution efficiency, and enhance customer satisfaction?

1. Executive Summary

This analysis evaluates the performance of a SaaS support operation using 100,000 support tickets covering the period from 2022 to 2025. The analysis focuses on ticket resolution, support workload, priority handling, reopening behavior, SLA performance, and customer satisfaction.

The analysis shows that the support operation has a 60.11% resolution rate, with 39,887 tickets remaining unresolved in the dataset. For completed tickets, the median resolution time is 29.91 hours, while the average is higher at 45.01 hours, indicating a noticeable long-resolution tail.

The support workload itself appears relatively stable over time. Annual ticket volume remained close to 25,000 tickets per year, increasing only slightly from 24,603 tickets in 2022 to 25,162 in 2025. This suggests that the unresolved workload is not simply the result of rapidly increasing ticket volume.

Resolution performance is also broadly consistent across support channels, issue types, customer segments, and SLA plans. This means there is no single channel or ticket category that clearly explains the overall resolution bottleneck.

Customer satisfaction presents a separate area of concern. The average CSAT among respondents is 3.20 out of 5, with 41.28% satisfied, 29.19% neutral, and 29.53% dissatisfied. Interestingly, the statistical analysis found almost no relationship between resolution time and CSAT, with a Spearman correlation of 0.002 and a p-value of 0.70875.

Overall, the analysis indicates that improving support performance requires two parallel approaches: improving operational control over unresolved and long-running tickets, and investigating customer experience factors beyond resolution speed, such as response quality, communication, and resolution effectiveness.

2. Dataset Overview

The dataset contains 100,000 support tickets and 20 fields covering operational and customer-experience information.

- Customer segment
- Support channel
- Product area
- Issue type
- Ticket priority
- Ticket status
- SLA plan
- Resolution time
- Reopen status
- Customer sentiment

- CSAT score
- Platform
- Region

The dataset contains no duplicate records. Resolution information is available for 60,113 tickets, while 39,887 tickets do not have a recorded resolution time. The region field contains some missing values, so regional analysis should be interpreted with caution.

3. KPIs

KPI	Result
Total tickets	100,000
Completed tickets	60,113
Unresolved tickets	39,887
Resolution rate	60.11%
Reopen rate	5.04%
CSAT responses	70,059
CSAT response rate	70.06%
Average CSAT	3.20 / 5
Median resolution time	29.91 hours
Average resolution time	45.01 hours

These KPIs provide the overall picture of the support operation. The most noticeable operational metric is the 39.89% unresolved rate. This represents a significant volume of tickets without a recorded resolution time and should be a major focus for support operations management.

4. Support Workload Analysis

Stable Ticket Volume

Year	Tickets
2022	24,603
2023	25,165
2024	25,070
2025	25,162

Ticket volume increased by only about 2.3% between 2022 and 2025. Quarterly volumes were also relatively balanced.

Quarter	Tickets
Q1	24,698
Q2	24,860
Q3	25,276
Q4	25,166

There is no strong indication of a major seasonal or yearly increase in support demand. This suggests that the unresolved-ticket problem cannot simply be explained by rapidly increasing ticket volume. The support team should therefore investigate internal factors such as ticket aging, workload distribution, escalation processes, and unresolved-ticket management.

5. Ticket Resolution Performance

Among the 60,113 completed tickets, resolution time shows a clear long tail.

Resolution Metric	Hours
Average	45.01
Median	29.91
75th percentile	51.66
90th percentile	108.77
95th percentile	175.19
Maximum	239.96

The difference between the median and average is significant. A typical completed ticket is resolved in around 30 hours, but a smaller group of tickets takes several days to resolve. The top 10% of completed tickets take more than approximately 108.77 hours to resolve.

This indicates that the support operation has a long-resolution tail. Management should monitor median resolution time, P90 resolution time, P95 resolution time, unresolved tickets, escalated tickets, and tickets approaching SLA limits.

6. Priority-Based Resolution

Priority	Mean Resolution	Median Resolution	P90
Urgent	26.71 hrs	4.91 hrs	115.68 hrs
High	31.56 hrs	14.75 hrs	106.15 hrs
Medium	43.28 hrs	29.80 hrs	106.45 hrs
Low	55.61 hrs	45.47 hrs	110.65 hrs

Urgent tickets have a median resolution time of only 4.91 hours, compared with 45.47 hours for low-priority tickets. The data indicates that high-priority tickets are receiving faster attention. However, low-priority tickets have a much longer typical resolution time, creating a potential backlog-management issue.

7. Resolution by Issue Type

Issue Type	Median Resolution
Feature request	30.49 hrs

How-to	30.16 hrs
Billing problem	30.06 hrs
Account access	30.00 hrs
Security concern	29.93 hrs
Performance	29.68 hrs
Other	29.54 hrs
Bug	29.48 hrs

The difference between the highest and lowest median is approximately 1 hour. There is no strong evidence that one particular issue type is creating the overall resolution bottleneck. Support improvement should therefore focus on broader workflow and backlog management.

8. Resolution by Product Area

Product Area	P90 Resolution	Long-Resolution Rate
Mobile app	118.10 hrs	10.87%
API integration	113.28 hrs	10.46%
Analytics dashboard	109.94 hrs	10.06%
Notifications	103.55 hrs	9.70%
Login/auth	104.53 hrs	9.69%
Billing	103.91 hrs	9.60%
Data export	102.98 hrs	9.59%

Mobile app tickets show a somewhat higher long-resolution rate. However, the difference is not large enough to identify the mobile app as the primary operational bottleneck. It should instead be treated as an area for monitoring, particularly when tickets remain open for several days.

9. Resolution by Support Channel

Channel	Median Resolution
Phone transcript	30.46 hrs
Chat	30.01 hrs
In-app	30.00 hrs
Web form	29.65 hrs
Email	29.57 hrs

There is no strong evidence that the support channel itself is responsible for slower resolution. Simply moving customers from one support channel to another is therefore unlikely to address the main operational issue identified in this analysis.

10. Customer Segment Analysis

Customer Segment	Median Resolution	P90 Resolution
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Small business	30.53 hrs	107.42 hrs
Enterprise	30.06 hrs	116.03 hrs
Individual	29.90 hrs	100.25 hrs
Non-profit	29.86 hrs	111.44 hrs
Education	29.18 hrs	108.01 hrs

Enterprise tickets have a P90 resolution time of 116.03 hours, compared with 100.25 hours for individual customers. This represents approximately a 15.7% difference in the long-resolution tail. Enterprise tickets may therefore require additional monitoring when they become long-running, although the typical resolution time is similar across segments.

11. Reopen Analysis

The overall reopen rate is 5.04%. Reopen rates remain relatively stable across issue types, with the highest issue-level rate for performance tickets at 5.26% and the lowest for how-to tickets at 4.81%. Product-level reopen rates range from approximately 4.65% to 5.22%, while priority-level rates range from approximately 5.00% to 5.46%.

Reopening does not appear to be strongly concentrated in one particular issue type, product area, or priority. It should therefore be treated as an overall quality metric rather than a problem limited to one category.

12. Customer Satisfaction Analysis

Customer satisfaction was available for 70,059 responses, representing a 70.06% response rate. The average CSAT score is 3.20 out of 5.

CSAT Category	Responses	Percentage
Satisfied	28,920	41.28%
Neutral	20,453	29.19%
Dissatisfied	20,686	29.53%

Approximately 58.72% of respondents were either neutral or dissatisfied. Customer satisfaction therefore has meaningful room for improvement.

13. Resolution Time vs Customer Satisfaction

A statistical test was performed to understand whether longer resolution times are associated with lower CSAT.

Statistical Measure	Result
Spearman correlation	0.002
P-value	0.70875

The correlation is effectively zero, and the p-value is well above the commonly used 0.05 threshold. CSAT also remains very similar across different resolution-time bands.

Resolution time does not show a meaningful statistical relationship with CSAT. Faster resolution is valuable for operational efficiency, but the data does not show that faster resolution by itself will necessarily produce higher customer satisfaction.

Other factors should be investigated, including quality of the solution, agent communication, accuracy of responses, first-contact resolution, customer expectations, number of interactions required, and whether the issue was actually solved. These factors are not fully available in the current analysis, so their impact cannot be quantified from this dataset.

14. Reopened Tickets and CSAT

Ticket Status	Average CSAT	Dissatisfied Rate
Not reopened	3.20	29.54%
Reopened	3.18	30.68%

Reopened tickets show slightly lower CSAT, but the difference is not large enough in this analysis to identify reopening as a major standalone driver of customer dissatisfaction.

15. SLA Performance

SLA Plan	Mean Resolution	Median	P90
Gold	45.14 hrs	29.89 hrs	110.23 hrs
Platinum	44.24 hrs	29.35 hrs	102.90 hrs
Standard	45.08 hrs	29.99 hrs	109.31 hrs

A Kruskal-Wallis test produced a p-value of 0.52916. This does not provide statistically significant evidence of a difference in resolution time across the SLA plans. CSAT is also very similar across the plans.

16. Key Findings

Unresolved workload is significant: 39,887 out of 100,000 tickets are unresolved, giving a resolution rate of 60.11%.

Long-running tickets create an efficiency challenge: Median completed-ticket resolution time is 29.91 hours, while P90 reaches 108.77 hours and P95 reaches 175.19 hours.

Priority handling is differentiated: Urgent tickets have a median resolution time of 4.91 hours, compared with 45.47 hours for low-priority tickets.

No single channel or issue type dominates the bottleneck: Resolution times are broadly similar across channels and issue types.

Customer satisfaction needs attention: Average CSAT is 3.20/5, with 29.53% of respondents dissatisfied.

Faster resolution alone may not improve CSAT: The Spearman correlation between resolution time and CSAT is only 0.002.

17. Business Recommendations

Create an Aging-Based Backlog Management Process

Make the unresolved ticket rate a core operational KPI. Group tickets by age, such as less than 24 hours, 24–48 hours, 48–72 hours, 72–120 hours, and more than 120 hours. Older tickets should trigger additional review or escalation.

Protect Low-Priority Tickets From Becoming Stale

Low-priority tickets have a median resolution time of 45.47 hours. Introduce queue-management rules so that low-priority tickets are periodically reviewed based on age, not only priority.

19. Final Business Outcome

The analysis shows that the SaaS support operation is not facing one isolated bottleneck. Instead, the main challenge is the combination of a large unresolved workload and a long tail of tickets requiring several days to resolve.

At the same time, resolution performance is relatively consistent across channels, issue types, customer segments, and SLA plans. This suggests that improvement efforts should focus on ticket aging, backlog management, escalation workflows, and operational visibility rather than simply changing the support channel or SLA structure.

Customer satisfaction requires a parallel approach. With an average CSAT of 3.20/5 and almost 30% of respondents dissatisfied, there is a clear opportunity to improve the customer experience. However, the statistical analysis shows that resolution time alone is not a meaningful predictor of CSAT in this dataset.

The most practical approach is therefore to manage support performance through two connected but separate goals: improve operational efficiency by reducing unresolved and long-running tickets, while improving customer experience by understanding the quality and effectiveness of the support interaction.